



National Teachers College

629 J Nepomuceno, Quiapo, Manila, 1001 Metro Manila
Bachelor of Science in Information Technology

ITELECTIVE 1 for Sustainable Development Goal

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I. Introduction:

Project Overview:

For this project, We developed an Employee Management System using VB.NET Windows Forms Application. The purpose of this system is to make attendance tracking easier and more organized by recording employee Time-In and Time-Out information digitally.

The system allows employees to log in using their Employee ID and password before accessing the attendance features. After logging in, employees can record their attendance and view their attendance history. The system also includes a report generation feature using Microsoft Report Viewer.

To store attendance information permanently, I used Microsoft SQL Server together with ADO.NET for database connection and data management.

This project supports Sustainable Development Goal (SDG) 8: Decent Work and Economic Growth because it helps improve workplace monitoring and attendance management.

Objectives

- **General Objective**

To create an Attendance Monitoring System that records employee attendance digitally and generates reports automatically.

- **Specific Objectives**

- To create a login system for employees.
- To record employee Time-In and Time-Out.
- To store attendance information using SQL Server.
- To display attendance records inside the application.
- To generate attendance reports using Report Viewer.

II. Requirements Analysis



Functional Requirements

- FR1 – Login Authentication-The system should allow employees to log in using Employee ID and Password.
- FR2 – Attendance Recording-The system should allow employees to perform Time-In and Time-Out.
- FR3 – Attendance Monitoring-The system should display attendance history inside the application.
- FR4 – Report Generation-The system should generate attendance reports using Microsoft Report Viewer.
- FR5 – Data Storage-The system should save attendance records in SQL Server database.
- FR6 – Input Validation-The system should validate empty fields before processing data.

These requirements follow the mandatory project specifications requiring CRUD/database persistence, reporting, validation, and login implementation.

Non-Functional Requirements

- NFR1 – UI Consistency-The interface should follow proper naming conventions and maintain a simple and organized design so users can easily use the system.
- NFR2 – Maintainability-The code should remain organized and easy to modify for future improvements.
- NFR3 – Reliability-The system should save attendance records correctly without losing information.
- NFR4 – Performance-Attendance records and reports should load without unnecessary delay.
- NFR5 – Security-The system should only allow authorized users to access attendance features.
- NFR6 – Database Connection Management-The system should maintain stable database connectivity using SQL Server and [ADO.NET](#).



III. Database Design

Data Dictionary

- **Table: Employees**

Field Name	Data Type	Constraint
EmpID	VARCHAR(50)	Primary Key
FullName	VARCHAR(100)	Not Null

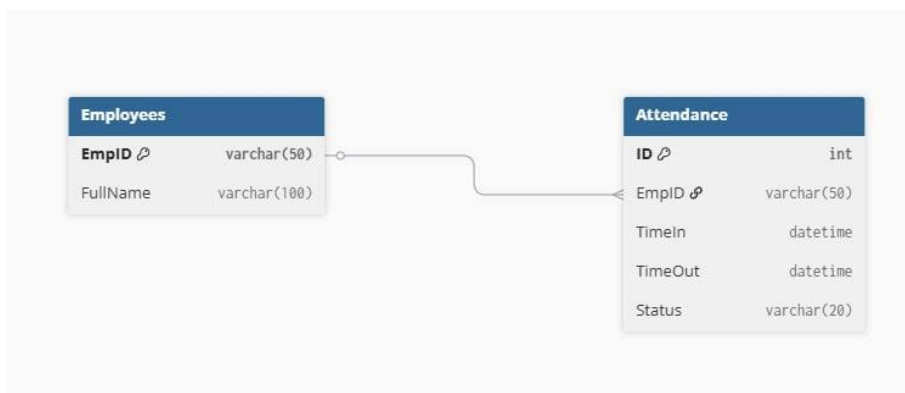
- **Table: Attendance**

Field Name	Data Type	Constraint
ID	INT	Primary Key Identity
EmpID	VARCHAR(50)	Foreign Key
TimeIn	DATETIME	Null
TimeOut	DATETIME	Null
Status	VARCHAR(20)	Not Null

Relationship Description

The Employees table has a one-to-many relationship with Attendance.

One employee can have multiple attendance records, while one attendance record belongs to only one employee.





IV. System Architecture

System Process Flow and Workflow Description

The AlgoFlow Employee Management System follows a structured and secure operational sequence that begins immediately at the application startup stage. Upon opening, the system loads the initial Login Form (Form1), automatically masking the password characters to protect credential privacy. The core process flow is split into two primary operational phases: User Authentication and Transactional Attendance Logging.

1. User Authentication Phase

Before entering the dashboard, the system enforces initial data validation. When a user inputs their Employee ID and password, the system checks for empty inputs. If any fields are left blank, a warning prompt stating "Please fill in all required fields" is displayed, returning the user to the input stage. Once validation passes, the system performs a password check against database records. If the password validation fails, a "Wrong Password!" message box appears, prompting a re-entry. Upon a successful match, the system securely stores the authenticated EmployeeID in memory, hides the login interface, and dynamically transitions to the main dashboard (Form2).

2. Transactional Attendance Logging and Reporting Phase

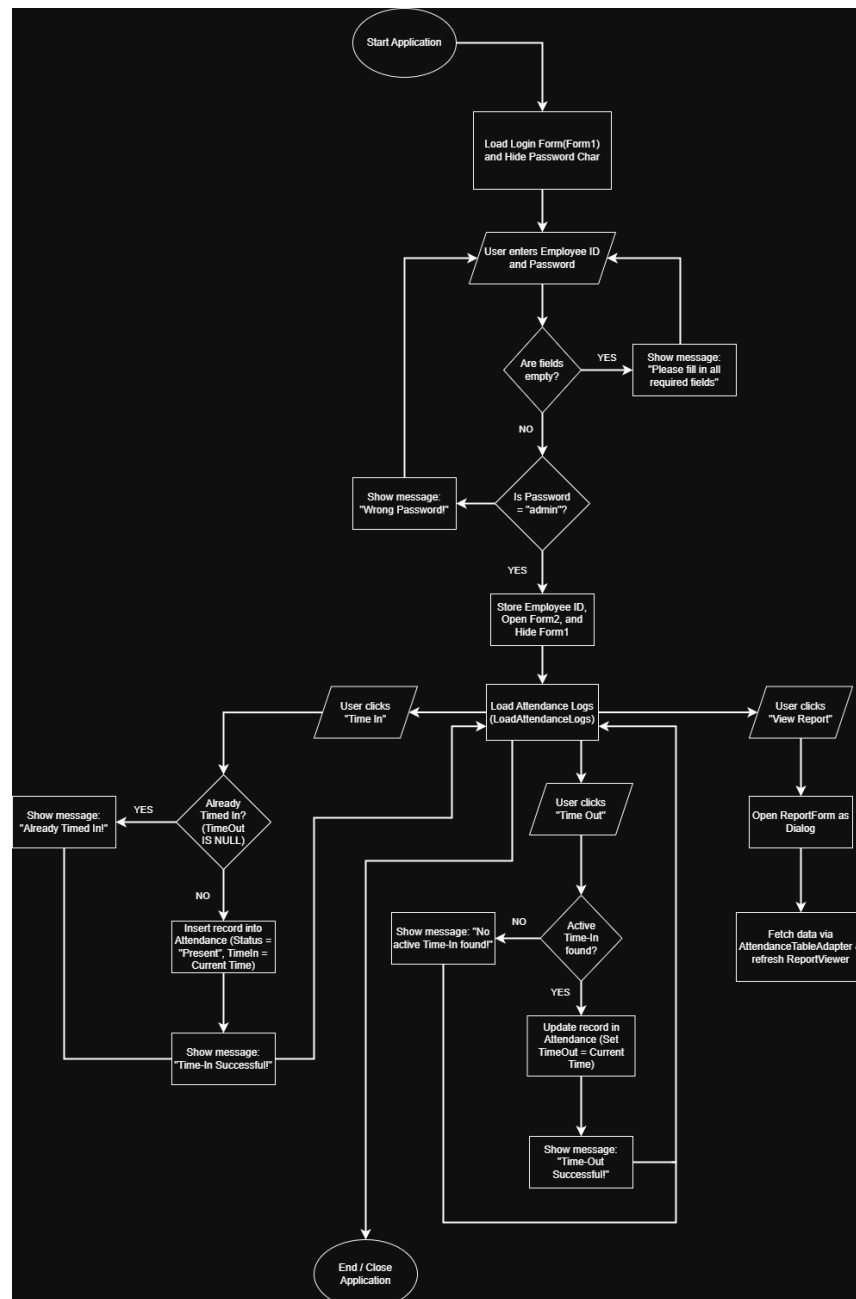
Once inside the main dashboard, the application immediately triggers the LoadAttendanceLogs procedure to populate the real-time status matrix. From this centralized cockpit, three independent, user-driven action paths can be taken:

- **The Time-In Process:** When the user clicks the "Time In" button, the system evaluates the database to check if the user has an active, unclosed shift (where TimeOut IS NULL). If an open shift is detected, it blocks the transaction and issues an "Already Timed In!" warning prompt. If no active session exists, the application executes a SQL database insert command—setting the entry record status to 'Present' and logging the TimeIn parameter as the current system time—before displaying a "Time-In Successful!" confirmation and refreshing the data grid logs.
- **The Time-Out Process:** When the user clicks the "Time Out" button, the system scans the historical ledger for an existing, active shift. If no matching open log is found, it prevents further execution and displays a "No active Time-In found!" message box. If an active session is validated, the application updates the specific database record, writing the exact current time into the TimeOut column field, followed by a "Time-Out Successful!" message before refreshing the view grid.



- The Report Generation Process: When the user selects the "View Report" action control, the system opens the dedicated ReportForm modal dialog window. This process immediately executes a data fetch routine utilizing the AttendanceTableAdapter, populating the embedded Microsoft ReportViewer control, and rendering an audit-ready summary of tracking metrics. The entire workflow safely loops back to the main dashboard tracking log state after completing any of these actions, allowing continuous usage until a user executes the close process to safely shut down database connectivity and trigger the End / Close Application state.

Flowchart





V. Technical Implementation

Tools and Technologies Used

Programming Language: VB.NET

Database: Microsoft SQL Server

Technologies:

- Windows Forms
- ADO.NET
- SQL Server
- Microsoft Report Viewer
- DataGridView

The system uses SQL Server together with ADO.NET for database connection and data handling. Report Viewer is used to generate attendance reports automatically.

VI. Code Explanation

- **Login Module**
 - The login form checks if Employee ID and Password fields are complete before allowing access.
 - If the password entered matches the expected value, the system opens the main attendance form.
 - Otherwise, an error message appears.
- **Attendance Module**
 - The Time-In function checks first if an employee already has an active attendance record.
 - If no active record exists, attendance information is saved into the database.
 - The Time-Out function updates the current attendance record by adding the employee's Time-Out value.
- **Report Module**
 - The report module retrieves attendance information from SQL Server and displays it using Microsoft Report Viewer.
 - This allows users to monitor attendance records more efficiently.



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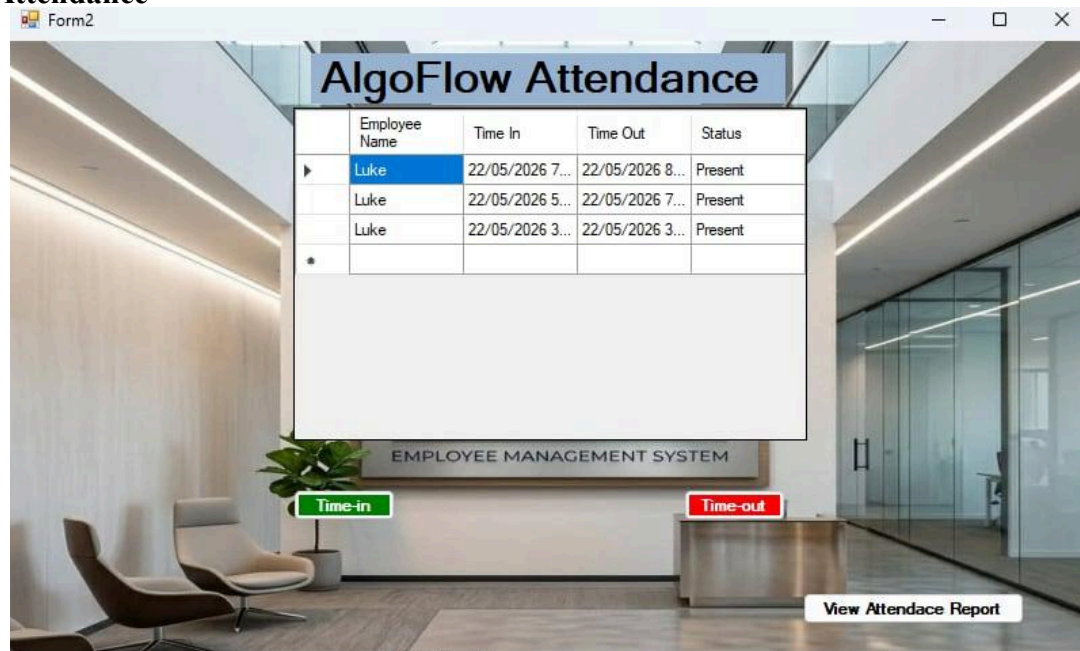
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VII. Testing and Evaluation

- Log-in



- Attendance





● Report

5/22/2026 3:44:44 PM5/22/2026 3:44:44 PM	5/22/2026 3:44:48 PM5/22/2026 3:44:48 PM	Present Present
5/22/2026 3:45:57 PM5/22/2026 3:45:57 PM	5/22/2026 3:46:02 PM5/22/2026 3:46:02 PM	Present Present
5/22/2026 5:29:38 PM5/22/2026 5:29:38 PM	5/22/2026 7:49:09 PM5/22/2026 7:49:09 PM	Present Present
5/22/2026 7:57:38 PM5/22/2026 7:57:38 PM	5/22/2026 8:43:34 PM5/22/2026 8:43:34 PM	Present Present

VIII. Conclusion

Developing this Attendance Monitoring System helped me understand how VB.NET, SQL Server, and ADO.NET work together in building a functional application.

Through this project, I learned how to create a login system, connect an application to a database, manage attendance records, and generate reports. The system provides a more organized and efficient way of monitoring employee attendance while reducing manual work.

This project also allowed me to apply programming concepts into a real-world situation and improve my skills in application development.

IX. Reference

- Chatgpt: Explained the activity and corrected the grammar.
- Gemini: Debugged our code because it had many errors.